Affine Transformations

Affine Transformation

An **affine transformation** is a geometric transformation that preserves **collinearity** (points lying on a straight line remain aligned) and **ratios of distances** (midpoints stay midpoints). It includes operations like **Scaling** and **Shearing.**

Midpoint Preservation (Collinearity + Ratios)

- If point M is the midpoint of points A and B **before** the transformation, then after applying an affine transformation:
- The *transformed* point M' will **still be the midpoint** of A' and B'

Example:

- Let $A = (0,0)$, $B = (4,0)$ and midpoint $M = (2,0)$
- Apply scaling by factor 2:
  - $A' = (0,0)$, $B' = (8,0)$, $M' = (4,0)$.
  - Observe: M' is *still* the midpoint of A' and B' (now at 4, halfway between 0 and 8).

Scaling

Scaling is a linear transformation that changes the size of an object by stretching or compressing it along the coordinate axes. It is represented by a **scaling matrix** that multiplies the coordinates of the object.

Mathematical Definition

Given a scaling factor s_x along the $x-axis$ and s_y along the $y-axis$, the scaling transformation is defined as:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$(x, y) \rightarrow$ Original Point
$(x', y') \rightarrow$ Transformed Point
$s_x \rightarrow$ Scaling factor in the x-direction
$s_y \rightarrow$ Scaling factor in the y-direction

- **Uniform Scaling:** $s_x = s_y$ (preserves proportions).
- **Non-Uniform Scaling:** $s_x \neq s_y$ (distorts the object).
- **Reflection:** If s_x or s_y is negative, the object flips.
- If Only Along $x-axis$ factor 2 then $s_x = 2, s_y = 1$

Example : Scaling of a triangle

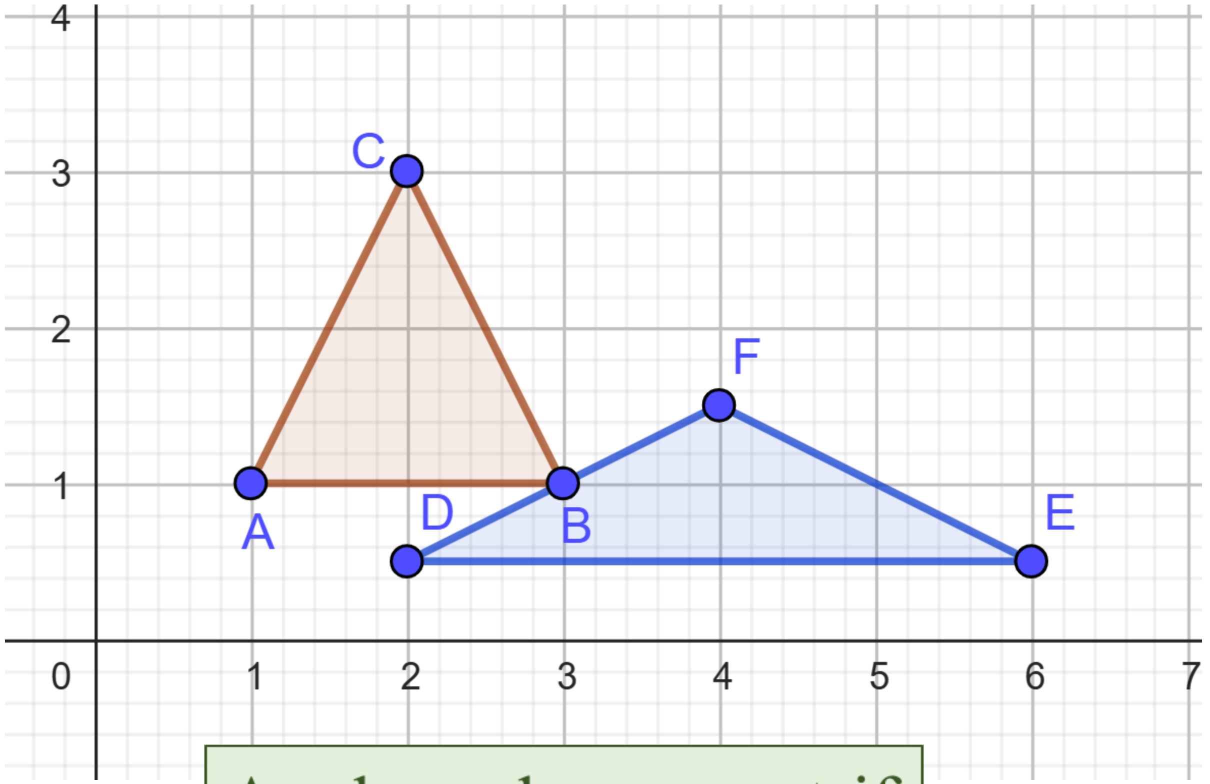
A triangle with vertices $A(1,1), B(3,1), C(2,3)$ with $s_x = 2,\ s_y = 0.5$

$$As: T(x) = A * x$$

$$T\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}2 & 0\\0 & 0.5\end{bmatrix} * \begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}2\\0.5\end{bmatrix}$$

$$T\begin{bmatrix}3\\1\end{bmatrix} = \begin{bmatrix}2 & 0\\0 & 0.5\end{bmatrix} * \begin{bmatrix}3\\1\end{bmatrix} = \begin{bmatrix}6\\0.5\end{bmatrix}$$

$$T\begin{bmatrix}2\\3\end{bmatrix} = \begin{bmatrix}2 & 0\\0 & 0.5\end{bmatrix} * \begin{bmatrix}2\\3\end{bmatrix} = \begin{bmatrix}4\\1.5\end{bmatrix}$$



Original Vertices: $A(1,1), B(3,1), C(2,3)$

Transformed Vertices: $D(2, 0.5), E(6, 0.5), F(4, 0.5)$

Apply and comment if

i. $s_x = 3, s_y = 3$

ii. $s_x = -2, s_y = 0.5$

iii. $s_x = s_y = -1$

Example: Scaling of a Circle

Determine the image of the circle $(x-2)^2 + (y-3)^2 = 4$ under the scaling by factor 2.

$$As: T(x) = A * x$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} * \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

$$x' = 2x \Rightarrow x = \frac{x'}{2}$$

$$y' = 2y \Rightarrow y = \frac{y'}{2}$$

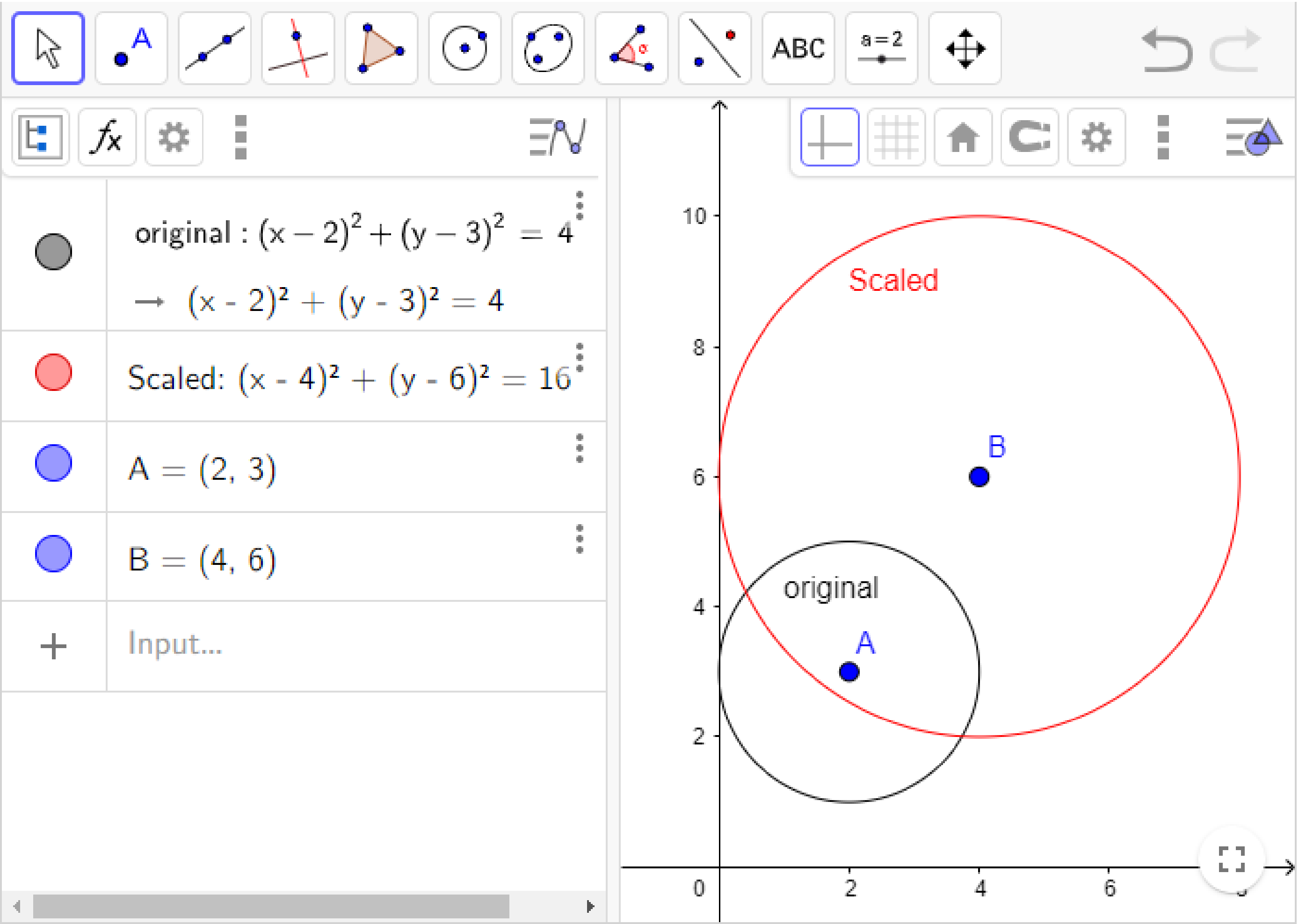
Putting in given eq.

$$\left(\frac{x'}{2} - 2\right)^2 + \left(\frac{y'}{2} - 3\right)^2 = 4$$

$$\frac{(x'-4)^2}{4} + \frac{(y'-6)^2}{4} = 4$$

$$(x'-4)^2 + (y'-6)^2 = 16$$

Transformed Circle



Scaling of Circle: Interesting Fact

If I apply a scaling transformation to a circle, sometimes its center also changes. Why does that happen?

- By default, in linear algebra and most software (like GeoGebra), scaling is applied **relative to the origin**.
- If the circle is **not centered at the origin**, scaling **will move its center**.
- Let's say a circle has center at $(2, 3)$ and radius 2
- If you scale by $s_x = 2, s_y = 2$, the new center will be $(4, 6)$ (scaled away from the origin).

Example: Scaling of Ellipse

Determine the image of the circle $\frac{x^2}{4^2} + \frac{y^2}{2^2} = 1$ under the scaling by factor 2.

$$As: T(x) = A * x$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} * \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

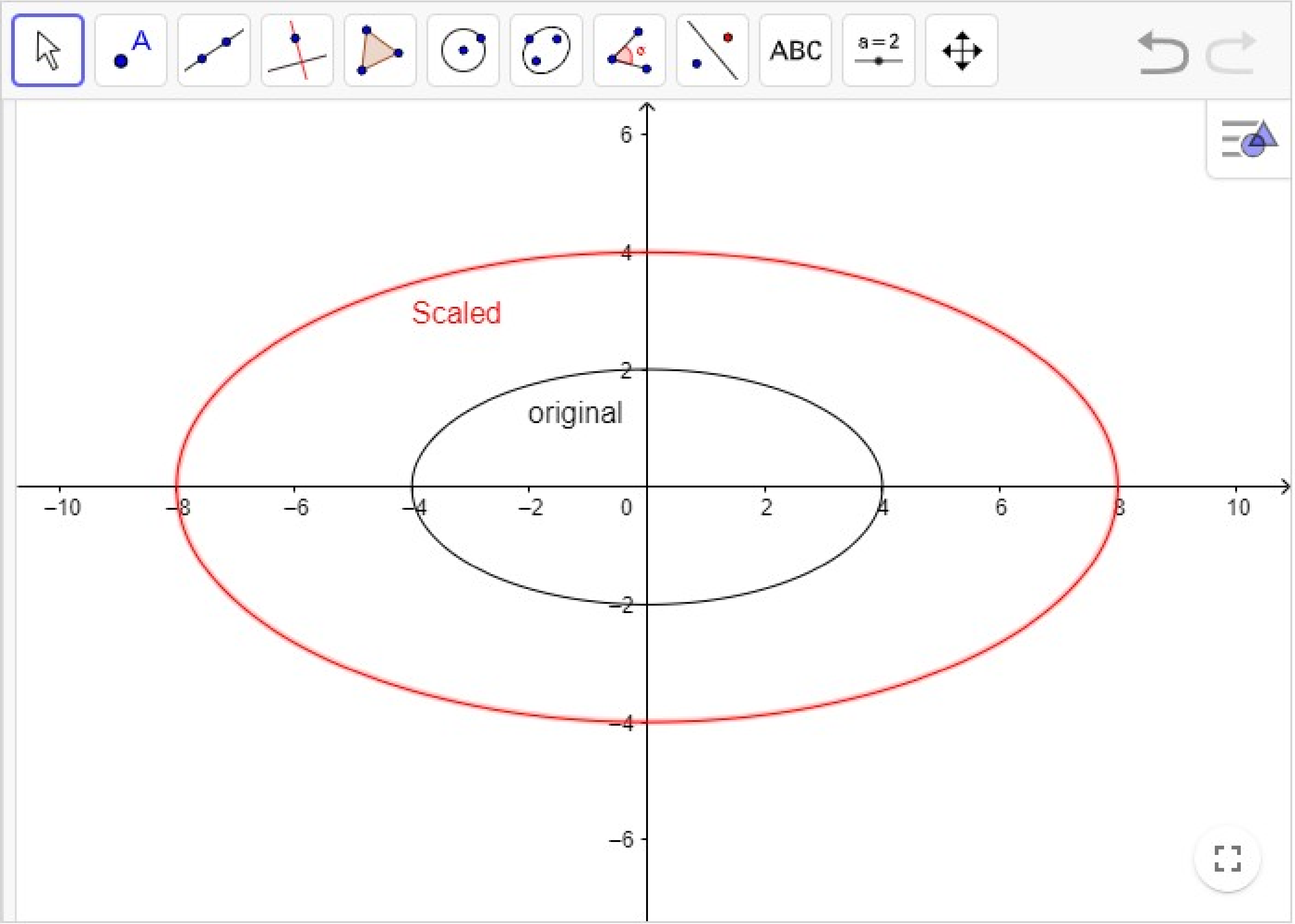
$$x' = 2x \Rightarrow x = \frac{x'}{2}$$

$$y' = 2y \Rightarrow y = \frac{y'}{2}$$

Putting in given eq.

$$\frac{1}{16}\left(\frac{x'}{2}\right)^2 + \frac{1}{4}\left(\frac{y'}{2}\right)^2 = 1$$

$$\frac{x'^2}{64} + \frac{y'^2}{16} = 1$$



Scaling: Applications in Computer Science

- **Computer Graphics:** Resizing images, zooming, 3D modeling.
- **Game Development:** Scaling sprites, UI elements.
- **Data Normalization:** Feature scaling in machine learning.

Scaling: Summary

- Scaling changes the size of an object using a diagonal matrix.
- Default scaling is relative to the origin.
- Uniform scaling preserves shape, while non-uniform scaling distorts it.

Practice Problems

Q1. A square with vertices $(0,0), (0,1), (1,0), (1,1)$. Scale this square by factor 2.

Q2. Find the Image of line$y = 2x + 5$ under the transformation of scaling by factor 3.

Q3. Find the image of an ellipse

$$\frac{(x-2)^2}{4} + \frac{(y-3)^2}{9} = 1$$

under the transformation of scaling by factor 2.

Q4. Let $(x-2)^2 + (y-1)^2 = 4$ be a circle. Find its equation and image under stretching along Y-axis by factor $\frac{3}{2}$.

Q5. Find the stretching of an ellipse $\frac{(x-4)^2}{16} + \frac{(y-1)^2}{4} = 1$ along x-axis by factor 2.

Shearing

Shear (or *skew*) is a linear transformation that displaces points along one axis proportionally to their coordinate on another axis, distorting the shape while preserving area. It "tilts" objects, turning squares into parallelograms and circles into ellipses.

Mathematical Definition

There are two primary types of shear in 2D:

X-direction Shear (Horizontal Shear)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & k_x \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Y-direction Shear (Vertical Shear)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ k_y & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

General Form (Combined Shear)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & k_x \\ k_y & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Example : Horizontal Shear

A Square with vertices $A(0,0), B(1,0), C(1,1)$, D(0,1) with $k_x = 1.5$

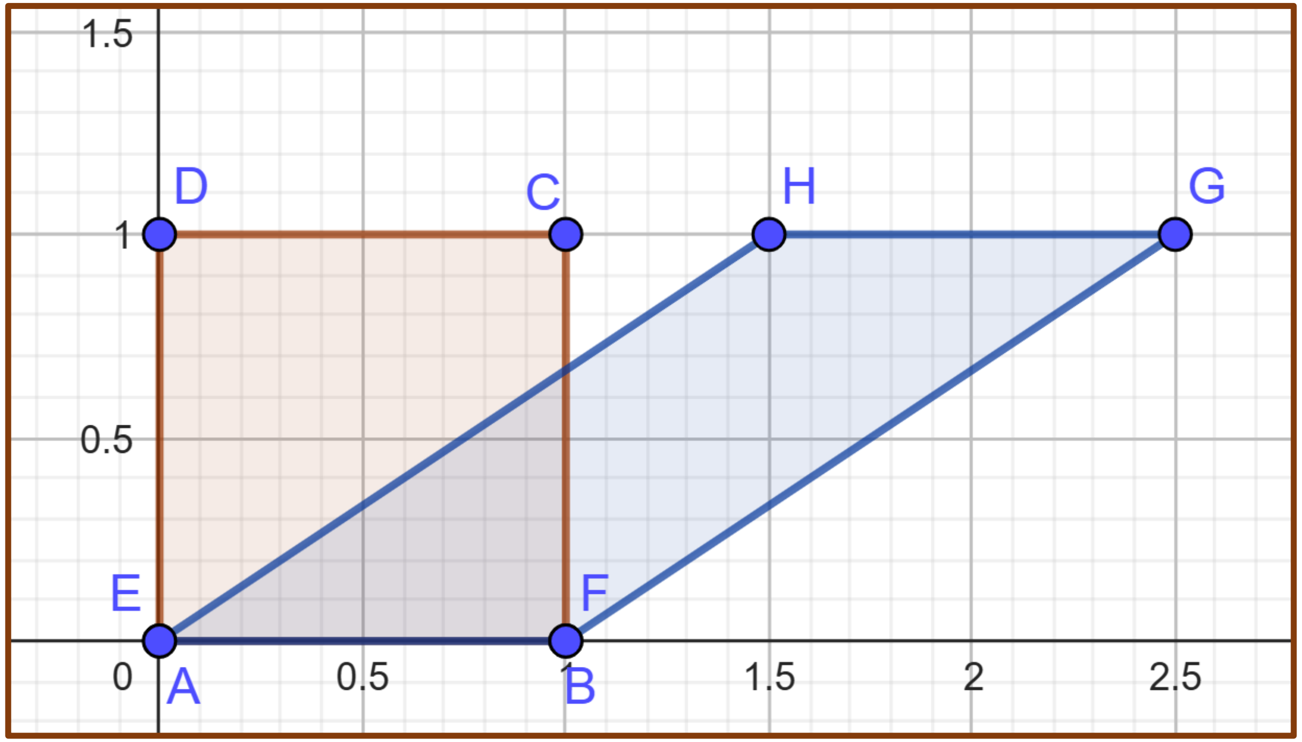
$$As: T(x) = A * x$$

$$T\begin{bmatrix}0\\0\end{bmatrix} = \begin{bmatrix}1 & 1.5\\0 & 1\end{bmatrix} * \begin{bmatrix}0\\0\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$$

$$T\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}1 & 1.5\\0 & 1\end{bmatrix} * \begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}1\\0\end{bmatrix}$$

$$T\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}1 & 1.5\\0 & 1\end{bmatrix} * \begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}2.5\\1\end{bmatrix}$$

$$T\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}1 & 1.5\\0 & 1\end{bmatrix} * \begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}1.5\\1\end{bmatrix}$$



Original Vertices: $A(0,0), B(1,0), C(1,1)$, D(0,1)

Transformed Vertices: E$(0,0), F(1,0), G(2.5,1), H(1.5,1)$

Example : Vertical Shear

A Square with vertices $A(0,0), B(1,0), C(1,1)$, D(0,1) with $k_y = 0.5$

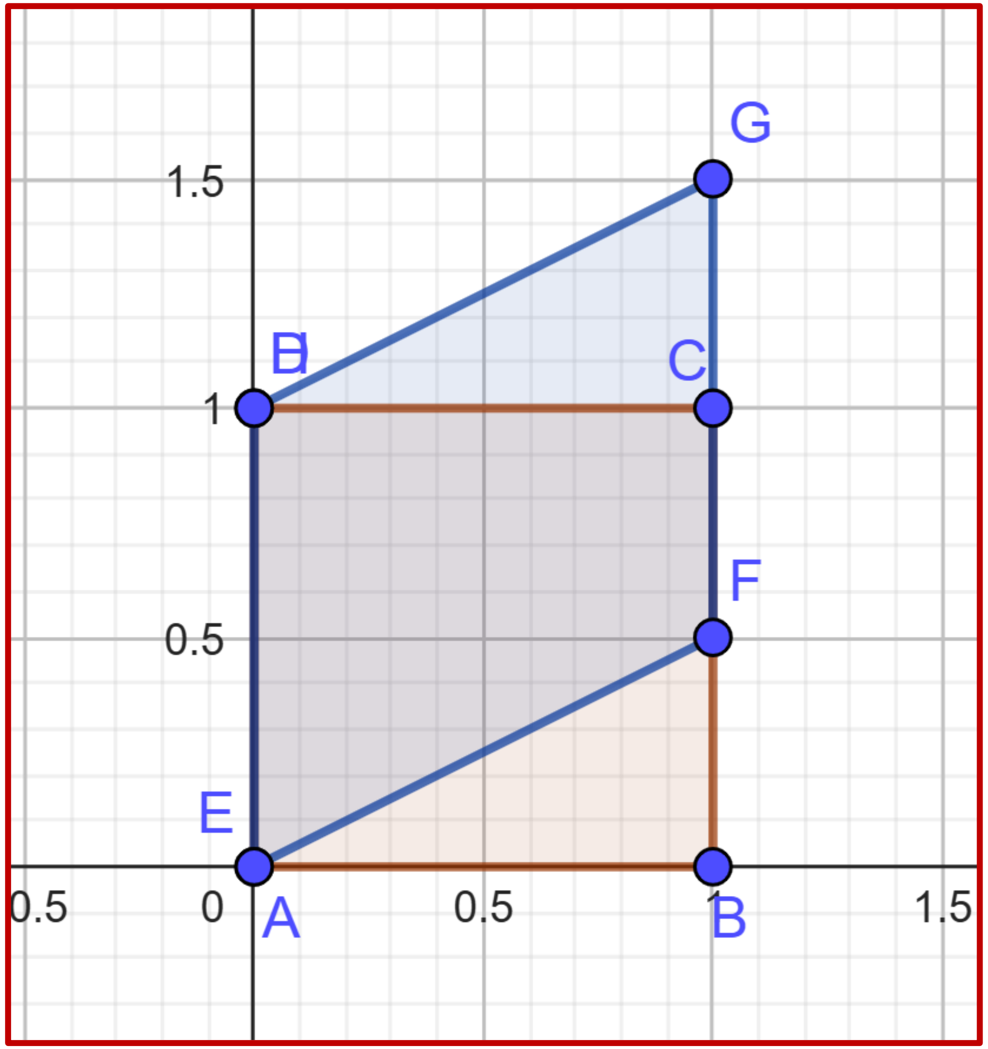
$$As: T(x) = A * x$$

$$T\begin{bmatrix}0\\0\end{bmatrix} = \begin{bmatrix}1 & 0\\0.5 & 1\end{bmatrix} * \begin{bmatrix}0\\0\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$$

$$T\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}1 & 0\\0.5 & 1\end{bmatrix} * \begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}1\\0.5\end{bmatrix}$$

$$T\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}1 & 0\\0.5 & 1\end{bmatrix} * \begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}1\\1.5\end{bmatrix}$$

$$T\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}1 & 0\\0.5 & 1\end{bmatrix} * \begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}0\\1\end{bmatrix}$$



Original Vertices: $A(0,0), B(1,0), C(1,1)$, D(0,1)

Transformed Vertices: E$(0,0), F(1,0.5), G(1,1.5), H(0,1)$

Example : Combined Shear

A Square with vertices $A(0,0), B(1,0), C(1,1)$, D(0,1) with $k_x = 0.5, k_y = 1$

$$As: T(x) = A * x$$

$$T\begin{bmatrix}0\\0\end{bmatrix} = \begin{bmatrix}1 & 0.5\\1 & 1\end{bmatrix} * \begin{bmatrix}0\\0\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$$

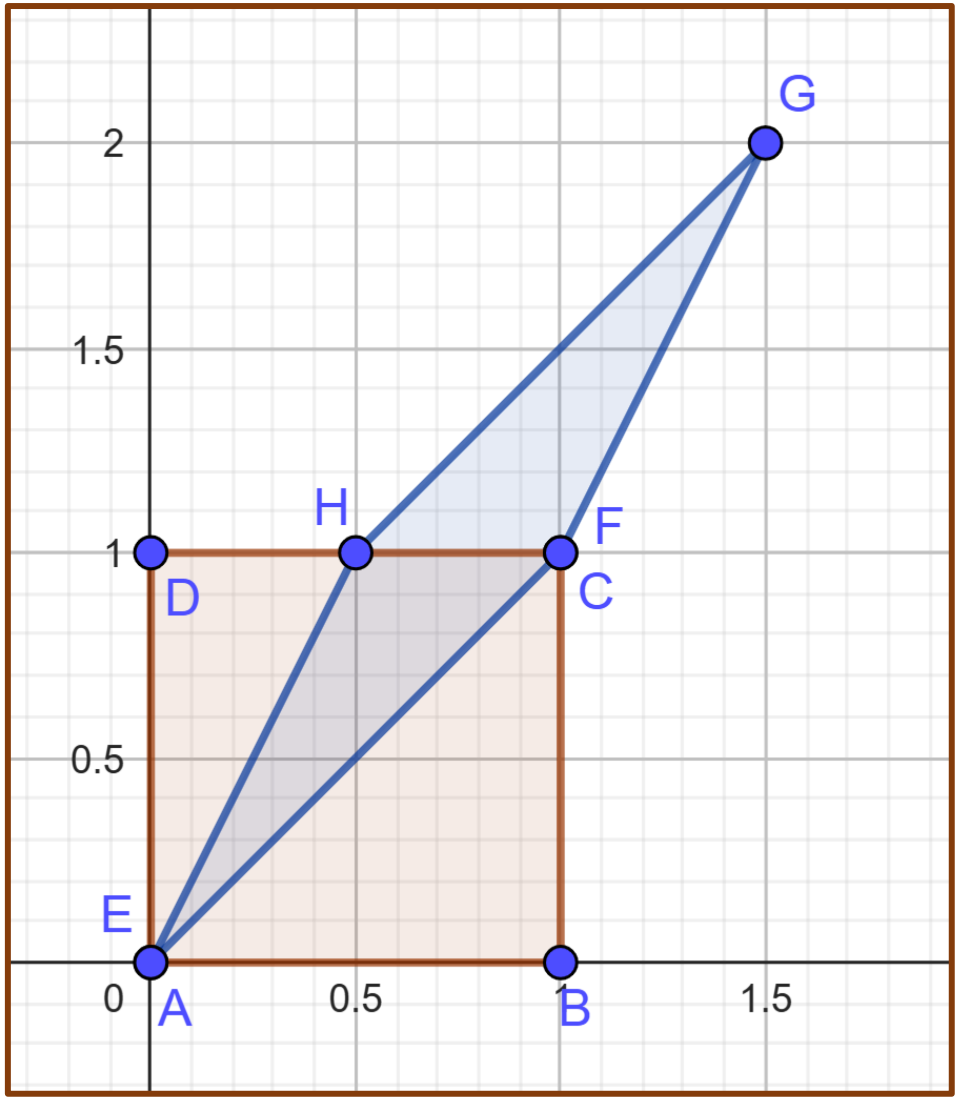
$$T\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}1 & 0.5\\1 & 1\end{bmatrix} * \begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}$$

$$T\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}1 & 0.5\\1 & 1\end{bmatrix} * \begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}1.5\\2\end{bmatrix}$$

$$T\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}1 & 0.5\\1 & 1\end{bmatrix} * \begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}0.5\\1\end{bmatrix}$$

Original Vertices: $A(0,0), B(1,0), C(1,1)$, D(0,1)

Transformed Vertices: E(0,0), $F(1,1), G(1.5,2), H(0.5,1)$



Example: Shearing of a Circle

Determine the image of the circle $x^2 + y^2 = 4$ under the effect of horizontal shear by a factor -2.

2nd Degree Equation $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ represents

i. Ellipse if $B^2 - 4AC < 0$

ii. Parabola if $B^2 - 4AC = 0$

iii. Hyperbola if $B^2 - 4AC > 0$

as, $2^2 - 4(1)(5) < 0$

So sheared equation of Circle is Ellipse.

$$As: T(x) = A * x$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x - 2y \\ y \end{bmatrix}$$

$x' = x - 2y$ $\quad$ $and\ y' = y$

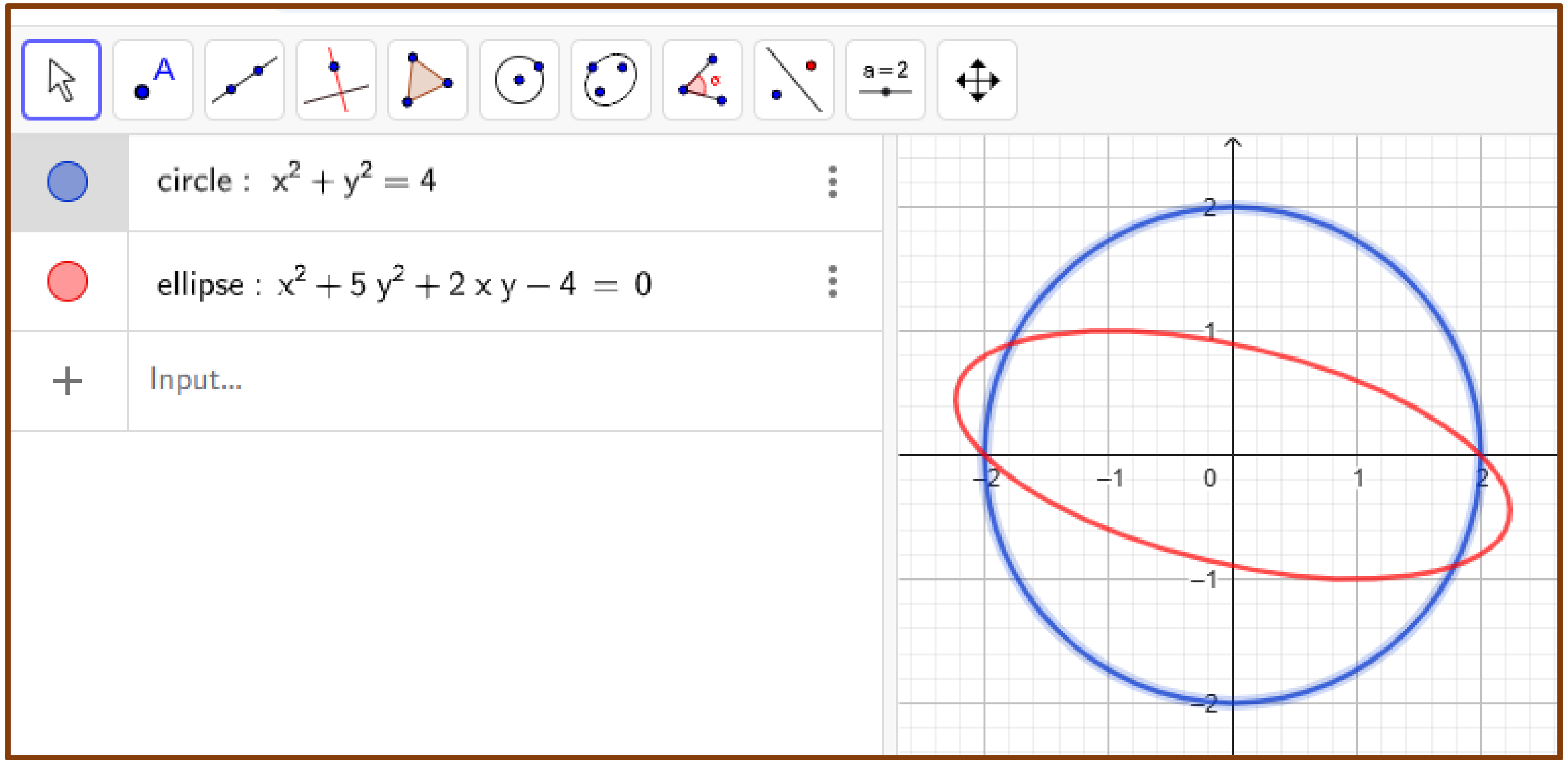
$so, x = x' + 2y'$

Putting in given eq.

$$(x' + 2y')^2 + y'^2 = 4$$

$$x'^2 + 4y'^2 + 2x'y' + y'^2 = 4$$

$$x'^2 + 2x'y' + 5y'^2 - 4 = 0$$



Shearing: Applications in Computer Science

- **Typography:** Slanted fonts (e.g., *italic* text).
- **Computer Graphics:** Simulating shadows, reflections, or deformations.
- **Physics:** Modeling shear stress in materials.

Practice Problems

Q1. Let $(x+2)^2+(y+1)^2 = 4$ be a circle. Find its equation and image under the effect of shear parallel to y-axis by factor $\frac{3}{2}$.

Q2. Let $(x-2)^2+(y-1)^2 = 4$ be a circle. Find its equation and image under the effect of shear parallel to x-axis by factor 2.

Q3. Consider the rectangle with vertices $(0,0), (0,2), (4,0), (4,2)$. Shear this rectangle along y-axis by factor -3.

Q4. In each part describe the matrix operator corresponding to A and show it effect on unit square

$a)\ \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix},$ b) $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix},$ $c)\ \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

Home Work:

Exercise in Word File

11. Affine Transformation

THANKS